

CFD 2025-26 – Lab 07

Consider a stationary Navier-Stokes problem in the pipe Ω of Figure 1:

$$\begin{cases} (\mathbf{u} \cdot \nabla) \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = 4y(1-y)\mathbf{i}, & \text{on } \Gamma_{\text{in}} := \{x=0, 0 \leq y \leq 1\}, \\ (\nu \nabla \mathbf{u} - p\mathbf{I})\mathbf{n} = \mathbf{0}, & \text{on } \Gamma_{\text{out}} := \{2 \leq x \leq 3, y=3\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Gamma_{\text{wall}} := \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}), \end{cases}$$

where the kinematic viscosity ν is equal to 10^{-2} and the boundary conditions are given by:

- no-slip boundary conditions on the lateral boundary Γ_{wall} of the pipe;
- a parabolic profile of velocity on the inlet boundary Γ_{in} ;
- homogeneous Neumann boundary conditions on the outlet boundary Γ_{out} .

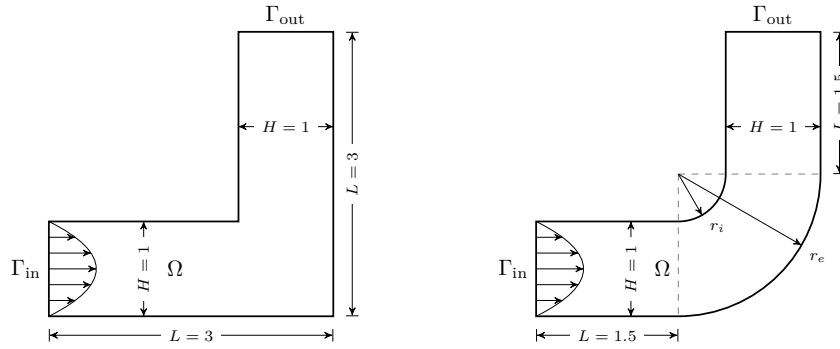


Figure 1: Domain Ω for Exercises 1 and 2 : miter joint (left: point 1, elbow1.msh) and 90° circular elbow (right: point 2, elbow2.msh).

Exercise 1

Let us take into account the following fixed-point iteration method: given $(\mathbf{u}_{k-1}, p_{k-1})$, the next iteration $(\mathbf{u}_k, p_k)$ is given by the solution of the linear problem

$$\begin{cases} (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{u}_k - \nu \Delta \mathbf{u}_k + \nabla p_k = \mathbf{0}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u}_k = 0, & \text{in } \Omega, \\ \mathbf{u}_k = 4y(1-y)\mathbf{i}, & \text{on } \Gamma_{\text{in}}, \\ (\nu \nabla \mathbf{u}_k - p_k \mathbf{I})\mathbf{n} = \mathbf{0}, & \text{on } \Gamma_{\text{out}}, \\ \mathbf{u}_k = \mathbf{0}, & \text{on } \Gamma_{\text{wall}} = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}). \end{cases}$$

- 1 Using the fixed-point iteration method defined above, solve the nonlinear problem over the geometry in Figure 1, left. As an initial condition $(\mathbf{u}_0, p_0)$, use the solution of the

corresponding Stokes problem. Employ a stopping criterion based on the increment:

$$\frac{\|\mathbf{u}_k - \mathbf{u}_{k-1}\|_{H^1}}{\|\mathbf{u}_{k-1}\|_{H^1}} + \frac{\|p_k - p_{k-1}\|_{L^2}}{\|p_{k-1}\|_{L^2}} \leq 10^{-3}.$$

- 2 Repeat the exercise using the smooth elbow geometry displayed in Figure 1, right. What differences can you observe?

Exercise 2 - homework

Let us take into account the Newton method, defined by the following iterative algorithm: given $(\mathbf{u}_{k-1}, p_{k-1})$, the next iteration $(\mathbf{u}_k, p_k)$ is given by the solution of the linear problem

$$\begin{cases} (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{u}_k + (\mathbf{u}_k \cdot \nabla) \mathbf{u}_{k-1} - \nu \Delta \mathbf{u}_k + \nabla p_k = (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{u}_{k-1}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u}_k = 0, & \text{in } \Omega, \\ \mathbf{u}_k = 4y(1-y)\mathbf{i}, & \text{on } \Gamma_{\text{in}}, \\ (\nu \nabla \mathbf{u}_k - p_k \mathbf{I}) \mathbf{n} = \mathbf{0}, & \text{on } \Gamma_{\text{out}}, \\ \mathbf{u}_k = \mathbf{0}, & \text{on } \Gamma_{\text{wall}} = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}}). \end{cases}$$

- 1 Using the Newton method defined above, solve the nonlinear problem, over the domain displayed in Figure 1, left. As an initial condition $(\mathbf{u}_0, p_0)$, use the solution of the corresponding Stokes problem. Employ a stopping criterion based on the increment:

$$\frac{\|\mathbf{u}_k - \mathbf{u}_{k-1}\|_{H^1}}{\|\mathbf{u}_{k-1}\|_{H^1}} + \frac{\|p_k - p_{k-1}\|_{L^2}}{\|p_{k-1}\|_{L^2}} \leq 10^{-3}.$$

- 2 Repeat the exercise using the smooth elbow geometry displayed in Figure 1, right. What differences can you observe?